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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program: B.Tech/MBATech Stream : Data Science | Year: II Semester: III

Subject: Managing Uncertainty | Time: 3 hrs (11:00am to 2:00pm)

Date: 14 / 06 / 2023 . | No. of Pages: 7

Marks: 100

Special Re-Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	20
CO- 1; SO-1 ; BL-1	A	The mean of the following frequency distribution was found to be 1.46 *No. of Accidents* — *No. of Days (frequency)* 0 — 46 1 — ? 2 — ? 3 — 25 4 — 10 5 — 5 Total — 200 days Calculate the missing frequencies.	5

<table>
<tr><td>CO-2 ;
SO-2 ;
BL-2</td><td>B</td><td>A brokerage survey reports that 30 percent of individual investors have used a discount broker, that is, one which does not charge the full commission. In a random sample of 9 individuals, what is the probability that
I. Exactly two of the sampled individuals have used a discount broker
II. Not more than three have used a discount broker
III. At least three of them have used a discount broker</td><td>2
2
1</td></tr>
<tr><td>CO-4 ;
SO-1 ;
BL-2</td><td>C</td><td>There are five machines in a factory. Of these machines, three are working properly and two are defective. Machines which are working properly produce articles each of which has independently a probability of 0.1 of being imperfect. For the defective machines this probability is 0.2. A machine is chosen at random and five articles produced by the machine are examined</td><td>5</td></tr>
<tr><td>CO- 5;
SO-2 ;
BL-1</td><td>D</td><td>In a small business firm, two typists are employed-typist A and typist B. Typist A types out, on an average, 30 pages per day with a standard deviation of 6.
Typist B, on an average, types out 45 pages with a standard deviation of 10. Which typist shows greater consistency in his output?</td><td>5</td></tr>
<tr><td>Q2</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q2
CO-1;
SO-1;
BL-1</td><td>A</td><td>Find the missing frequency in the following distribution, for which the median is 48.25 and mode, is 44.
<table>
<tr><td>Class Interval</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td><td>80-90</td><td>Total</td></tr>
<tr><td>Frequency</td><td>3</td><td>7</td><td>?</td><td>?</td><td>?</td><td>20</td><td>16</td><td>4</td><td>150</td></tr>
</table></td><td>10</td></tr>
<tr><td>Q2
CO-1;
SO-1;</td><td></td><td>Calculate the most suitable average for the following data:</td><td>10</td></tr>
</table>

<table>
<tr><td>BL-2</td><td>B</td><td><table><tr><td>Size of the Item</td><td>Below 50</td><td>50-100</td><td>100-150</td><td>150-200</td><td>200 and above</td></tr><tr><td>Frequency</td><td>15</td><td>20</td><td>36</td><td>40</td><td>10</td></tr></table></td><td></td></tr>
<tr><td>Q3</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q3
CO-2;
SO-2; BL-2</td><td>A</td><td>The following distribution relating to marks obtained by students in an examination:
Calculate Standard deviation of the marks
<table><tr><th>Marks</th><th>Number of Students</th></tr><tr><td>0- 10</td><td>1</td></tr><tr><td>10- 20</td><td>3</td></tr><tr><td>20- 30</td><td>6</td></tr><tr><td>30- 40</td><td>10</td></tr><tr><td>40- 50</td><td>12</td></tr><tr><td>50- 60</td><td>11</td></tr><tr><td>60- 70</td><td>6</td></tr><tr><td>70- 80</td><td>3</td></tr><tr><td>80- 90</td><td>2</td></tr><tr><td>90-100</td><td>1</td></tr></table></td><td>10</td></tr>
<tr><td>Q3
CO-2;
SO-1; BL-2</td><td>B</td><td>Calculate Karl pearson's coefficient of skewness.
<table><tr><th>Variable</th><th>Frequency</th><th>Variable</th><th>Frequency</th></tr><tr><td>70-80</td><td>11</td><td>30-40</td><td>21</td></tr><tr><td>60-70</td><td>22</td><td>20-30</td><td>11</td></tr><tr><td>50-60</td><td>30</td><td>10-20</td><td>6</td></tr><tr><td>40-50</td><td>35</td><td>0-10</td><td>5</td></tr></table></td><td>10</td></tr>
<tr><td>Q4</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q4
CO-;3
SO-2;
BL-1</td><td>A</td><td>Distribution of monthly income of a group of 3000 school teachers in a village conforms to a normal curve with $\mu = 600$ and $\sigma = 100$.
I. Find the percentage of teachers having monthly income of more than Rs. 800.
II. Number of teachers having monthly income less than Rs. 400.
III. Find the lowest monthly income amongst the highest paid 100 teachers.</td><td>10</td></tr>
</table>

		IV. The highest monthly income among the lowest paid teachers. V. Calculate the range of 3σ	
Q4 CO-;3 SO-2; BL-1	B	Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be I. 0 jobs; II. exactly 2 jobs; III. at most 3 arrivals. IV. What is the maximum jobs that should arrive one minute with 90 % certainty? V. more than 3 Jobs	10
Q5		Answer briefly:	20
Q5 CO-4; SO-2; BL-2	A	For an experiment of throwing a die twice, find the probability: I. of the event of getting a total of 9, given that the die has shown up points between 4 and 6 (both inclusive) II. of the event of getting points between 4 and 6 (both inclusive), given that a total of 9 has already been obtained	5 5
Q5 CO-4; SO-2; BL-2	B	An Economist believes that during periods of high economic growth, the Indian Rupee appreciates with probability 0.70; in periods of moderate economic growth, it appreciates with probability 0.40; and during periods of low economic growth, the Rupee appreciates with probability 0.20. During any period of time the probability of high economic growth is 0.30; the probability of moderate economic growth is 0.50 and the probability of low economic growth is 0.20. Suppose the Rupee value has been appreciating during the present period. What is the probability that	10

		we are experiencing the period of (I) high, (II) moderate, and (III) low, economic growth?	
Q6		Answer briefly:	20
Q6 CO-5; SO-2; BL-3	A	The random variable X takes values -1, 0, 1 with probabilities 1/8, 2/8, 5/8 respectively. I. Compute E(X). II. Plot the PMF III. Give the pmf of $Y = X^2$ and use it to compute E(Y). IV. Instead, compute $E(X^2)$ directly from an extended table. V. Compute Var(X).	10
Q6 CO-5; SO-2; BL-2	B	Suppose now that events A, B and C are mutually independent with P(A) = 0.3, P(B) = 0.4, P(C) = 0.5. Compute the following: (Hint: Use a Venn diagram) I. $P(A \cap B \cap C^c)$ II. $P(A \cap B^c \cap C)$ III. $P(A^c \cap B \cap C)$	3 3 4
Q7		Answer briefly:	20

<table>
<tr>
<td>Q7
CO-2;
SO-2;
BL-3</td>
<td>A</td>
<td>The following table shows the results of two colleges. Which is better on the average?
<table>
<tr><td>Colleges →</td><td colspan="2">College A</td><td colspan="2">College B</td></tr>
<tr><td>Courses ↓</td><td>No.of students</td><td>Marks %</td><td>No.of students</td><td>Marks %</td></tr>
<tr><td>I degree</td><td>200</td><td>70</td><td>150</td><td>80</td></tr>
<tr><td>II degree</td><td>150</td><td>60</td><td>100</td><td>60</td></tr>
<tr><td>III degree</td><td>100</td><td>80</td><td>50</td><td>80</td></tr>
</table>
</td>
<td>10</td>
</tr>
<tr>
<td>Q7
CO-3; SO-2;
BL-3</td>
<td>B</td>
<td>Explain Type I and Type II error</td>
<td>10</td>
</tr>
</table>

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986